

MULLINGAR, IrelandWMO: **039710**Lat: **53.533N**Lon: **7.367W**Elev: **100**StdP: **100.13**Time Zone: **1.00 (IRL)**Period: **98-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.9	-2.2	-5.3	2.4	-2.0	-3.8	2.8	-0.5	10.9	9.0	9.9	8.9	1.2	350	0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.6	22.6	17.2	21.0	16.4	19.6	15.6	18.1	21.0	17.2	19.7	16.4	18.7	3.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.0	12.3	19.3	16.1	11.6	18.4	15.3	11.0	17.5	51.4	21.1	48.6	19.7	46.3	18.8	22.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	8.9	7.9	7.0	-6.4	25.7	2.1	1.8	-7.9	26.9	-9.1	28.0	-10.3	29.0	-11.8	30.3	
(5)				-6.6	20.0	2.1	1.2	-8.1	20.8	-9.4	21.5	-10.5	22.1	-12.0	23.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.4	4.8	5.1	6.2	8.2	11.0	13.5	15.0	14.6	12.9	9.9	6.8	5.0
	DBStd	4.55	2.90	2.79	2.73	2.54	2.44	2.38	2.13	1.91	2.27	2.68	2.80	3.42
	HDD10.0	796	161	138	119	64	17	2	0	0	3	35	101	155
	HDD18.3	3248	418	371	375	304	229	147	105	116	163	261	346	412
	CDD10.0	594	1	1	3	11	47	106	156	143	90	32	4	1
	CDD18.3	4	0	0	0	0	0	1	3	1	0	0	0	0
	CDH23.3	36	0	0	0	0	1	9	22	4	0	0	0	0
	CDH26.7	2	0	0	0	0	0	1	1	1	0	0	0	0
Wind	WSAvg	3.4	3.8	3.7	3.6	3.5	3.4	3.0	2.8	2.9	3.1	3.4	3.5	3.8
Precipitation	PrecAvg	945	93	71	71	63	69	66	66	81	84	94	91	93
	PrecMax	1169	154	192	126	112	147	140	158	180	183	173	203	162
	PrecMin	647	28	9	14	20	12	19	15	7	2	35	35	20
	PrecStd	110	35	38	29	25	36	32	31	36	41	38	34	38
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.5	13.0	15.6	18.4	22.1	24.6	26.0	23.1	21.5	17.4	14.5	13.2
		MCWB	11.1	11.1	10.8	13.0	16.2	17.1	19.2	17.9	17.2	14.9	13.2	11.9
	2%	DB	11.2	11.6	13.1	16.1	19.7	21.6	23.1	21.0	19.6	15.9	13.1	11.9
		MCWB	10.2	10.1	10.0	11.5	14.6	16.1	17.6	17.0	16.1	13.9	12.0	11.0
	5%	DB	10.4	10.6	11.8	14.2	17.4	20.0	21.0	19.5	18.1	15.0	12.2	11.1
		MCWB	9.5	9.3	9.5	10.5	13.2	15.4	16.7	16.0	15.2	13.3	11.2	10.2
	10%	DB	9.5	9.7	10.7	12.7	15.7	18.3	19.3	18.4	16.9	14.2	11.3	10.1
		MCWB	8.6	8.5	8.8	9.8	12.2	14.4	15.6	15.4	14.5	12.6	10.3	9.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.4	11.5	11.8	13.7	17.0	18.2	19.8	19.1	18.2	15.7	13.6	12.2
		MCDB	12.0	12.4	14.1	17.6	21.4	22.1	24.9	21.5	20.2	16.7	14.1	12.8
	2%	WB	10.5	10.5	10.8	12.2	15.2	16.9	18.4	17.7	16.9	14.6	12.3	11.1
		MCDB	11.1	11.4	12.3	15.0	19.0	20.6	21.9	19.7	18.6	15.4	12.8	11.7
	5%	WB	9.6	9.6	10.0	11.1	13.9	15.8	17.2	16.9	15.8	13.7	11.4	10.3
		MCDB	10.3	10.5	11.5	13.5	16.5	19.1	19.9	18.7	17.5	14.7	12.1	11.0
	10%	WB	8.7	8.7	9.1	10.1	12.7	14.9	16.2	16.0	14.9	12.8	10.5	9.3
		MCDB	9.5	9.6	10.5	12.3	15.0	17.7	18.6	17.7	16.4	13.9	11.2	10.0
Mean Daily Temperature Range	5% DB	MDBR	5.4	6.1	7.2	7.9	8.1	8.1	7.6	7.5	7.4	6.8	5.9	5.5
		MCDBR	6.0	6.0	8.1	10.3	10.9	10.8	10.4	8.8	8.5	6.6	5.9	6.0
		MCWBR	5.7	5.1	5.6	6.4	6.6	6.2	5.8	5.3	5.6	5.1	5.3	5.7
	5% WB	MCDBR	5.9	5.9	7.2	9.1	9.5	9.7	9.0	7.6	7.4	5.9	5.7	5.9
MCWBR		5.8	5.5	5.6	6.0	6.2	5.9	5.7	5.3	5.5	5.1	5.5	5.8	
Clear-Sky Solar Irradiance	taub	0.332	0.342	0.365	0.375	0.384	0.373	0.376	0.358	0.352	0.340	0.326	0.322	
	taud	2.247	2.316	2.280	2.281	2.292	2.351	2.350	2.411	2.427	2.415	2.306	2.211	
	Ebn at Noon	612	748	810	853	863	876	866	863	821	745	620	537	
	Edh at Noon	64	81	103	117	122	116	115	102	89	73	59	55	
All-Sky Solar Radiation	RadAvg	0.69	1.39	2.47	3.78	4.79	4.96	4.49	3.77	2.83	1.68	0.88	0.53	
	RadStd	0.06	0.12	0.26	0.40	0.37	0.53	0.41	0.36	0.15	0.14	0.09	0.07	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (28 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page